



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGIES

SUMMER 2026

FINAL EXAMINATION

COURSE TITLE:	Relational Database and Web Integration
COURSE CODE:	SEN3102
CREDIT VALUE	6
INSTRUCTOR:	Engr. Tanwi Nkiamboh
DEPARTMENT	Software Engineering
EXAMINATION HALL	B.S. Chumbow Hall
DATE:	Monday, September 28, 2026
TIME:	8:00 - 10:30

INSTRUCTIONS:

- ✓ **ANSWER ALL THE QUESTIONS**
- ✓ **Comment at each step in writing your queries**
- ✓ **Your report shouldn't exceed 30 pages**
- ✓ **Credit is given for legibility, clarity of expressions and use of relevant illustrations.**
- ✓ **Clearly write your registration number on each answer sheet used**

Scenario

You are hired by ICTU Events Ltd to design and build a full-stack conference and ticketing management system that integrates a relational database with a secure web application. The system must manage attendees, speakers, events, bookings, sessions, and payments while ensuring both performance optimization and security.

Part 1 – Database Design & Enhancement (15 marks)

1. Draw a **complete ER diagram** including all tables shown in the dataset images below. (10 marks)
2. Suggest **two additional tables** that could improve the system functionality and justify their role. (5 marks)

Part 2 – Dataset (Images Provided)

Attendees Table

attendee_id	first_name	last_name	email	phone
1	John	Nangu	nangu@ict.edu.cm	678475601
2	Pascal	Cho	cho@ict.edu.cm	677710578
3	Winnie	Onari	onari@ict.edu.cm	690871033
4	Grace	Tima	tima@ict.edu.cm	655091237

Speakers Table

speaker_id	first_name	last_name	topic
1	Prof	Ngwa	AI in Education
2	Dr	Cheikh	Cybersecurity Trends
3	Engr	Nanga	Web App Optimization

Events Table

event_id	event_name	event_date	location	total_seats	ticket_price
1	Tech Conference 2025	2025-09-15	Main Hall	100	15000.0
2	Cybersecurity Workshop	2025-09-20	Lab A	50	20000.0
3	AI in Education Forum	2025-09-25	Auditorium	150	10000.0

Bookings Table

booking_id	attendee_id	event_id	seats_booked	amount_paid	booking_date
1	1	1	2	30000.0	2025-08-01
2	2	1	1	15000.0	2025-08-03
3	3	2	1	20000.0	2025-08-05
4	4	3	3	30000.0	2025-08-06

Sessions Table

session_id	event_id	speaker_id	session_title	start_time	end_time
1	1	1	AI Overview	09:00	10:30
2	1	3	Optimizing Web Apps	11:00	12:30
3	2	2	Cyber Threats 2025	10:00	12:00
4	3	1	AI in Classrooms	14:00	15:30

Part 3 – Relational Algebra Queries (10 marks)

Using the dataset above, write relational algebra expressions to:

1. Identify attendees who booked more than 2 seats for any single event. (2 marks)
2. Retrieve the list of events that are fully booked (no seats remaining). (2 marks)
3. List the speakers who are presenting at more than one event. (2 marks)
4. Identify attendees who have not booked any event. (2 marks)
5. List events where total revenue > 30,000. (2 marks)

Part 4 – SQL Queries (20 marks)

1. Insert a new booking for attendee **Alice Nzokou** for the **Cybersecurity Workshop** with 2 seats. (3 marks)
2. Update the ticket price of **AI in Education Forum** to 12,000. (3 marks)
3. Retrieve the total revenue per event in descending order. (3 marks)
4. List all sessions with event name, speaker full name, and session time. (3 marks)
5. Find the attendee who spent the most money on bookings. (4 marks)
6. Create a stored procedure to update available seats automatically when a booking is made. (4 marks)

Part 5 – Web Application Integration (15 marks)

1. Build a responsive event listing page that displays all events with available seats. (5 marks)
2. Implement a ticket booking form that is linked to the database. (5 marks)
3. Implement secure **user authentication**. (5 marks)

Part 6 – Optimization & Security (10 marks)

1. Create indexes to improve the performance of searches on event names and booking dates. (3 marks)
2. Use prepared statements to guard against SQL Injection attacks. (3 marks)
3. Explain **two performance optimization strategies** applied. (4 marks)

ALL THE BEST